Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which of the following best defines a pattern in itemset data? 1 / 1 point

- A superset of all items within a dataset.
- A sequence of numerical values representing transactions.
- A structure of attributes representing intrinsic properties of data objects.
- An arbitrary group of unrelated data objects.

Nice work

Correct! A pattern is defined as a structure that captures important characteristics of the data, and it can take the form of frequent subsets, association rules, or correlations.

2. What is an association rule in data mining? 1 / 1 point

- A rule that measures the association between two item sets.
- A rule that calculates the average support of item sets.
- A rule that predicts the next item in a sequence.
- A rule that identifies the most frequent item in a database.

Nice work

Correct! An association rule measures the relationship between two item sets.

3. What does the downward closure property state? 1 / 1 point

- Any frequent item set must have a frequent subset.
- Any subset of a frequent item set must also be frequent.
- All item sets in a database are frequent.
- Frequent item sets are always larger than their subsets.

Nice work

Correct! This is the definition of the downward closure property.

4. What principle does the Apriori algorithm follow to reduce the number of candidate item sets? 1 / 1 point

- If any item set is frequent, then its superset is frequent.
- If any item set is frequent, then its subset is infrequent.
- If any item set is infrequent, then its subset is also infrequent.
- If any item set is infrequent, then its superset is infrequent.

Nice work

Correct! The Apriori algorithm uses this principle to avoid checking supersets of infrequent item sets.

5. What is a closed pattern in the context of frequent itemset mining? 1 / 1 point

- An item set that is part of a larger pattern with higher support.
- An item set that is frequent and has no frequent super pattern.
- An item set that is infrequent but has a frequent subset.
- An item set that is frequent and has no super pattern with the same support.

Nice work

Correct! A closed pattern is an item set that is frequent and has no super pattern with the same support.

6. What is confidence in the context of association rules? 1 / 1 point

- The likelihood of an item set being a max pattern.
- The probability of an item set occurring in the dataset.
- The frequency of an item set in the entire dataset.
- The conditional probability of itemset Y given itemset X in the transactions.

Nice work

Correct! Confidence measures the conditional probability of Y given X in the transactions.

7. What does a lift score of 1.33 suggest about the association between two items? 1 / 1 point

- There is no association between the antecedent and the consequent.
- The antecedent is negatively associated with the consequent.
- The rule is not useful.
- The antecedent is positively associated with the absence of the consequent.

Nice work

Correct! A lift score greater than 1 suggests a positive association, and in this context, it indicates a positive association with the absence of the consequent.

8. What does a higher chi-square value indicate in the context of an association between two variables? 1 / 1 point

- A stronger association between the variables.
- No association between the variables.
- An error in the calculation of expected values.
- A weaker association between the variables.

Nice work

Correct! A higher chi-square value indicates a more significant association between the variables.

9. What does a higher mutual information value indicate about two variables? 1 / 1 point

- They have a stronger correlation and dependence.
- They are more independent of each other.
- They are unrelated to each other.
- They have less variability.

Nice work

Correct! A higher mutual information value indicates a stronger correlation and dependency between the variables.

10. How is mutual information applied in document classification? 1 / 1 point

- By identifying significant word pairs that contribute to the classification task.
- By analyzing the grammatical structure of sentences.
- By determining the frequency of each word in the document.
- By translating documents into a common language.

Nice work

Correct! Mutual information identifies significant word pairs that help in classifying documents.

11. What is the role of mutual information in language understanding? 1 / 1 point

- To translate languages into a common format.
- To identify significant relationships between words that aid comprehension.
- To determine the frequency of words in a language.
- To analyze the grammatical structure of language.

Nice work

Correct! Mutual information identifies significant relationships between words, aiding in language understanding.

12. What is a key challenge in mining frequent itemsets from large databases? 1 / 1 point

- Ensuring all patterns are correlated.
- Eliminating all frequent patterns.
- Avoiding the use of any algorithms.
- Finding patterns that meet certain statistical significance criteria.

Nice work

Correct! One of the challenges is finding patterns that meet certain statistical significance criteria.

13. What is the role of the Apriori algorithm in frequent itemset mining? 1 / 1 point

- To increase the number of item sets checked.
- To scale up frequent itemset mining by making the process more efficiently.
- To identifyeliminate all non-frequent item sets.
- To ensure all item sets are statistically significant.

Nice work

Correct! The Apriori algorithm helps scale up frequent itemset mining by reducing the number of item sets that need to be checked.

14. How is the similarity between two itemsets typically measured? 1 / 1 point

- By counting the number of items in each itemset.
- By comparing the length of each itemset.
- By evaluating the price of items in each itemset.
- By calculating the intersection and union of the itemsets.

Nice work

Correct! Similarity is often measured by looking at the intersection and union of itemsets.

15. What is the range of values for the Jaccard similarity index? 1 / 1 point

- -1 to 1
- 0 to 100
- 1 to 10
- 0 to 1

Nice work

Correct! The Jaccard similarity index ranges from 0 to 1, where 0 indicates no similarity, and 1 indicates identical sets.

16. What happens to the Jaccard similarity if the union of two sets is large (assuming the intersection of the two sets remain the same)? 1 / 1 point

- The similarity becomes zero.
- The similarity decreases.
- The similarity increases.
- The similarity remains unchanged.

Nice work

Correct! A larger union generally decreases the Jaccard similarity because the denominator in the similarity calculation increases.

17. How can the Jaccard similarity be used in social networks? 1 / 1 point

- To determine the most popular user.
- To calculate the number of followers.
- To block unwanted users.
- To recommend new friends based on shared connections.

Nice work

Correct! The Jaccard similarity can be used to recommend new friends by measuring similarity in friend lists.

18. What is the Jaccard similarity between two item sets with no common items? 1 / 1 point

- 2
- 0.5
- 1
- 0

Nice work

Correct! A Jaccard similarity is 0 when two item sets have no items in common.